

Comparative Analysis of ST-Based SRAM Cell Design in CMOS and CNFET Technologies

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Abstract—The growing demand for energy-efficient and reliable memory solutions in nanoscale systems necessitates innovation in Static Random-Access Memory (SRAM) cell design. Carbon Nanotube Field Effect Transistor (CNFET) are exceedingly well suited for high performance, low-power computing applications due to their superior electrical properties and nanoscale structure. This paper presents a comparative analysis of Schmitt Trigger (ST) based SRAM cell architectures implemented using both traditional CMOS and CNFET technologies. In this paper, analyzed four different types of SRAM cells 10T, 11T, 12T and 13T with ST on CMOS, CNFET technology to understand their effects and compare it. Furthermore, total Energy for 10T, 11T, 12T, and 13T of CNFET technology has been reduced by 75.15%, 74.66%, 82.52%, and 78.29%, respectively, as compared to CMOS technology. Moreover, the power consumption of 12T at CNFET is 1.52 μ W smaller than that of CMOS, which is 8.74 μ W. Write SNM (WSNM) reveals superior performance compared to 12T CMOS and CNFET, with values of 640.35mV and 615.28mV, respectively. In addition, Read SNM (RSNM) for 13T and 11T reaches 375.76mV and 336.71mV. Overall, 32nm CNFET technology showed the most favorable results. The simulations were done in Cadence Virtuoso with 32nm CMOS technology and Stanford CNFET model.

Keywords—Schmitt Trigger, SRAM, CNFET, CMOS, Power Consumption, Total Energy, Static Noise Margin (SNM), Delay.

I. INTRODUCTION

In present-day circumstances technological perspective, low-power applications and energy-efficient device growing needed for sustaining the rapid expansion of portable electronics, Internet of Things (IoT) systems, as well as high-performance computing. These devices, ranging from smartphones and wearable tech to smart appliances and autonomous systems, demand efficient memory and processing solutions to meet the high expectations for speed, reliability, and battery life. Semiconductor memory is a rudimentary part of these employed digital systems by seamlessly connecting data processing and storage, enabling fast access and low-latency operation that are essential for energy-efficient computing [1]. Among the various type of semiconductor memories cache memory is crucial because it

acts as a bridge linking the processor to the main memory [2]. SRAM is particularly well suited for this role, due to its high-speed operation, low latency and ability to retain data without refreshing [3], [4]. SRAM stores data using cross coupled inverters that can maintain their state indefinitely as long as power is supplied. CMOS technology-based SRAM design continues to dominate to maintain data stability across process variation and compatibility with submicron scaling. With continued scaling problems like short channel effect causes Drain Induced Barrier Lowering (DIBL) which in return increases leakage current resulting in higher static power, also degrade stability [5], [6], [7]. There are several approaches to achieve lower leakage current, lower power consumption with high stability for SRAM design, one effective method is Schmitt Trigger (ST). The ST is a regenerative circuit, act as a positive feedback, on account of this enables the circuit to high noise immunity as well as ensure soft-switching behavior during read and write operations [4]. CMOS technology have employed into implement ST-based SRAM cells. If CNFET technologies, which have enhanced I_{ON}/I_{OFF} and I_{ON}/C_{GG} ratios, are used in place of CMOS, it offer more power and energy-efficient methods for memory devices [8]. CNFETs have emerged to represent a promising next generation for SRAM design, due to its symmetrical behavior between p-type and n-type devices, high carrier mobility as well as balanced subthreshold electrical features, which attend to requirements for both low power consumption and high-performance systems [9]. Carbon nanotubes are made of graphene sheets by rolling into a cylindrical structure [10]. For this reason, it provides higher carrier mobility, better subthreshold slope and improve scalability which can even overcome the limitation of CMOS in the submicron scaling [11].

Numerous efforts to improve the architecture of the SRAM bit cell utilizing CMOS and CNFET technologies have been documented in previous studies. The author [5] proposed a SRAM design based on CNFET at the 5-nm technology node. It offers superior read, write, and static power efficiency than 7-nm Fin Field-Effect Transistor (FinFET) SRAM cell, but was not able to mention energy consumption. An 8T SRAM cell based on CNFET technology was proposed [12],

which was capable of minimizing the effects of process fluctuations. The cell demonstrates a reduction in dynamic power dissipation, exhibits minimal power consumption during hold mode, and enhances cell stability during RSNM. However, it has a smaller raise in read and write power. In paper [3] represent that CNFET based MT (Multi-Threshold) SRAM designs which focused on leakage reduction mechanisms was applied on conventional 6T, 7T, 8T and 9T. The proposed cells are capable of mitigating undesired effects, including short channel effects and mobility degradation. Authors [8] present various types design of ultra-low-power ST SRAM implementing on CNFET, also proposed a 6T simplified multi-threshold Voltage (V_{th}) 6T CNFET SRAM, and benchmarked it with 45nm CMOS technology. The power consumption of SRAM based on CNFET is lower than CMOS technology, along with a significantly reduced propagation delay, but energy consumption was not provided. Paper [13] proposed a single-ended ST10T SRAM cell, demonstrating reduced power consumption, faster read/write speeds, and improved stability over existing 7T, 8T, 9T, and 10T variants; however, the paper hadn't covered energy consumption metrics. Nevertheless, they did not reveal read stability of the cell. In [6] a modified ST-based single-ended 11T SRAM cell for ultra-low-voltage and low-power applications was demonstrated using HSPICE with 7-nm FinFET technology. Compared to 6T, 8T, ST10T, Modified-PMOS-PMOS-NMOS 10T (MPPN10T), Feedback Cutting 11T (FC11T), ST11T, and ST12T, the proposed SRAM cell improves leakage power consumption, enhances read and write static noise margins, and minimizes read and write delay but did not mention Hold SNM value. Authors [14] proposed a 13T cell with improved RSNM and WSNM which includes a power gating approach to enhance the write abilities of the cell. The proposed cell is constructed using 180 nm node and is evaluated against conventional 6T SRAM cell with respect to stability, power consumption, delay, and leakage power but write stability was not provided. In paper [7] a 10T single-ended SRAM cell was proposed with ST to enhance RSNM and reduce static power. The proposed cell compared to conventional 6T cell had lower static power, better read, write and hold stability. Nonetheless, energy consumption of the cell was not provided.

Considering the limitations of prior research, this work presents the first comprehensive analysis of ST based SRAM cells using both CMOS and CNFET technologies, highlighting their design trade-offs. This paper fills a crucial research gap by demonstrating the scalability and energy efficiency of CNFET based ST SRAM, establishing them as a viable option for future low-power memory applications.

II. SCHMITT TRIGGER SRAM CELLS

This section represents a brief overview of various existing ST-based SRAM cells and flow of operations. Figure 2, 4, 6 and 8 illustrates the WRITE, READ and HOLD operation of each SRAM cell in 32nm CNFET technology.

A. Schmitt trigger 10T SRAM Cell

A ST10T SRAM cell featuring a cross-coupled structure with a ST inverter (PM2, NM3, NM4, NM6), a stacked inverter (PM1, PM3, NM1, NM2), a pass gate (NG), and an access transistor (NM5), managed by row-based Word Line (WL), Write WL (WWL), and column-based signals Word Line A (WLA) and Word Line B (WLB) illustrated in Figure 1 [13].

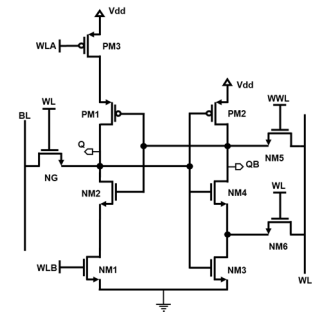


Fig. 1. Circuit diagram of ST10T SRAM cell [13].

During the Write '0' operation, Bit Line (BL) is driven low while WLB, WLA, WL, and WWL are kept high; PM3 is deactivated to power-gate node Q, and the pass gate NG drives Q to '0'. For Write '1', BL is high with WL and WWL on, WLA and WLB are kept low; Q flips from '0' to '1' via NG. In Read, BL is pre-charged to $V_{DD}/2$, WLA and WL are low, WLB is high, PM3 and NM1 are active and transistors NG and NM6 are inactive. If $Q = '0'$, QB becomes '1', turning on NM2 and discharging BL through NG and NM2–NM1. While in Hold, WLA, WL, and WWL are kept at low levels, WLB is high. Deactivate NG, NM5, and NM6, along with ensuring PM3 and NM1 remain active, isolating BL and retaining data. Figure 2 demonstrates WRITE, READ, and HOLD operations, including the pre-charge effect in the ST10T SRAM cell using 32nm CNFET technology.

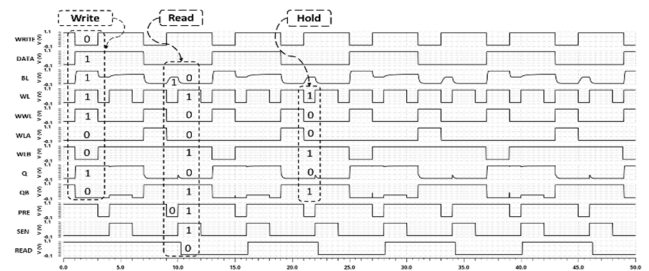


Fig. 2. WRITE, READ and HOLD operation for ST10T SRAM in 32nm CNFET technology.

B. Schmitt trigger 11T SRAM Cell

Figure 3 shows the modified ST11T SRAM cell, which uses two cross-coupled ST-based inverters [6]. The feedback-cutting transistor (NMF) and write access transistor (NMA1) are managed by the WL pulse, with BL connected to NMF via NMA1.

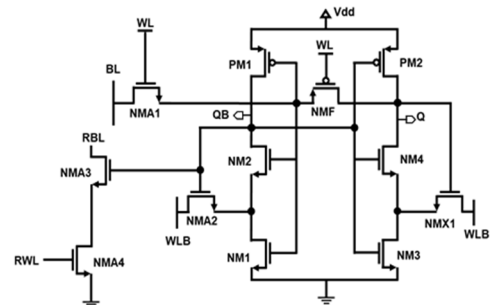


Fig. 3. Circuit Diagram of ST11T SRAM Cell [6].

The read circuit in ST11T cell uses NMA2 and NMA3, controlled by QB and Read Word Line (RWL), to discharge Read Bit Line (RBL) based on stored data. Transistor NMX1

and NMA2 have links to the ST inverters, which are regulated by the values stored in nodes Q and QB. For write operations, BL is set high/low for writing '1'/'0', WL is enabled, and WLB as well as RWL are disabled. Before reading, RBL is pre-charged to $V_{DD}/2$, WL is off, and RWL is turned on. If Q is '0' and QB is '1', NMA3 conducts, enabling RBL discharge through NMA4. In hold mode, WL and RWL remain low while WLB is active, preserving data in the storage nodes. Figure 4 shows WRITE, READ and HOLD operation of ST11T SRAM cell in 32nm CNFET technology.

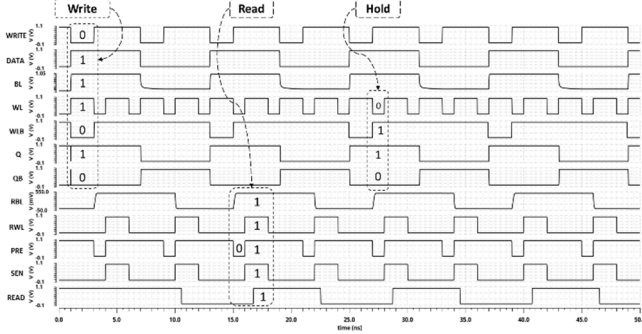


Fig. 4. WRITE, READ and HOLD operation for ST11T SRAM in 32nm CNFET technology.

C. Schmitt trigger 12T SRAM Cell

The architecture of a ST12T SRAM cell made up of two cross-coupled ST-based inverters shown in Figure 5. Two PMOS transistors, PM2 and PM4, assist the triggering of low voltage during write operation [15]. The WWL and WL is connected to transistor NML1, NMR1, PM2 and NML2, NMR2; respectively.

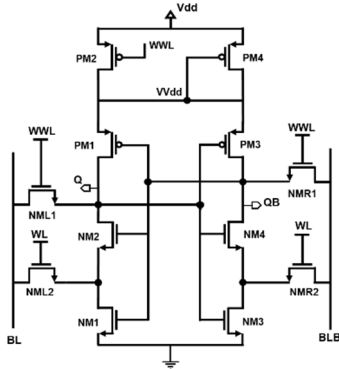


Fig. 5. Circuit Diagram of ST12T SRAM Cell [15].

When performing write operation, WWL is activated, which disables PM2 and disconnects the cell from the power supply. For a Write '0' operation, both WL and WWL are enabled, and BL and Bit Line Bar (BLB) are set to '0' and '1', respectively. The data is written to the storage nodes via NML1 and NMR1, setting Q to '0' and QB to '1'. Similarly, for a Write '1', BL and BLB are set to '1' and '0', respectively, resulting in Q = '1' and QB = '0'. During read operation, WWL is deactivated, enabling PM2 to reconnect V_{DD} to the cell. Half of the supply voltage ($V_{DD}/2$) is pre-charged into the bit lines before read. If Q = '1' and QB = '0', when WL is high then NML2 and NMR2 turn on, allowing V_{DD} to pass through PM2-PM1 and maintain Q at '1'. Meanwhile, QB is discharged through NM4-NM3, driving BLB to '0'. This voltage differential is detected by the sense amplifier to complete the read operation. Both WL and WWL

are disabled in hold mode, which separates the cell from the BL while preserving the data. Figure 6 shows WRITE, READ and HOLD operation of ST12T SRAM cell in 32nm CNFET technology.

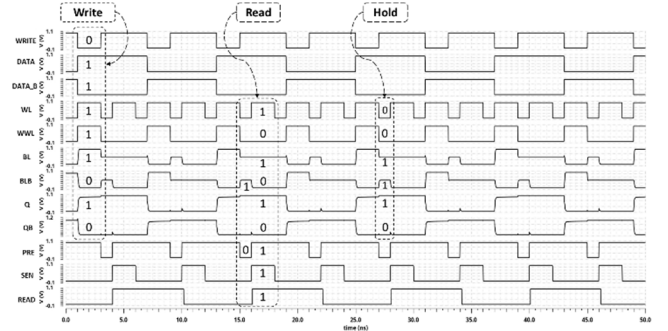


Fig. 6. WRITE, READ and HOLD operation for ST12T SRAM in 32nm CNFET technology.

D. Schmitt trigger 13T SRAM Cell

The ST13T SRAM cell, shown in Figure 7, is designed with two ST-based inverters and independent paths for read and write operations [14]. The right-side inverter consists of PM3, NM5, NM4, and NMX2, while the left-side inverter includes PM2, PM1, NM3, NM2, and NM1, with feedback provided by NMX1. The write access is managed by NMA1, connecting the cell to BL via WL. For read operations, NMA2 and NMA3, controlled by RWL and QB, respectively, RBL will be discharge through the path NMA2-NMA3.

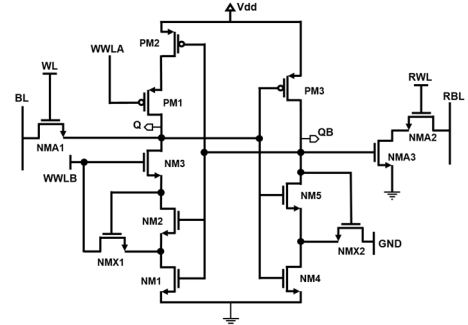


Fig. 7. Circuit Diagram of ST13T SRAM Cell [14].

To perform a write '0' operation, BL is set to '0', Write Word Line A (WWLA) is kept at a high level, and then PM1 is turned off, which terminates the connection of node Q from V_{DD} . Enabling Write Word Line B (WWLB) and WL allows QB to write '1' since latch Q is '0'. In a write '1' operation, BL is '1' and WL is enabled, and WWLB and WWLA is at lower level turning off NM3 and turning on PM1 respectively this changes help write '1' is Q node and QB node to '0'. Reading begins by pre-charging RBL to $V_{DD}/2$ before enabling RWL, which triggers transistor NMA3, while the control signals WWLA, WL is set to '0' and WWLB '1'. If node Q is '0' and QB '1', transistor NMA3 activates, thus giving a discharging path for RBL through the NMA2-NMA3. In Hold operation, WL, WWLA is at lower level and WWLB is on then the data can be retained in Q and QB node. Additionally, A major change can be seen if feedback transistor NMX2 is not GND, it will generate massive leakage current through NM5 and NM4 which in return increase static power consumption. Figure 8 illustrate WRITE, READ and HOLD operation of ST13T SRAM cell 32nm CNFET technology.

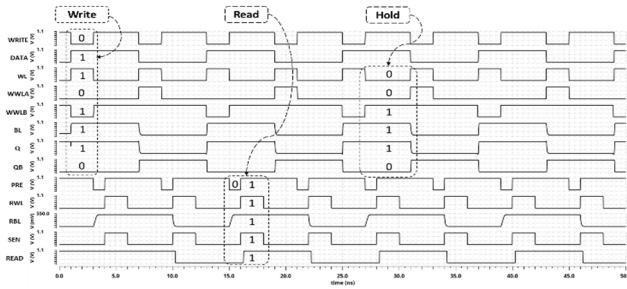


Fig. 8. WRITE, READ and HOLD operation for ST13T SRAM in 32nm CNFET technology.

III. PERFORMANCE EVALUATION

This section provides a comparative evaluation of various SRAM cell designs implemented in two different technologies in Cadence Virtuoso to perform key parameters such as power consumption, static power, delay, SNM and total energy consumption.

A. Power Consumption

SRAM bit cells have two different forms of power consumption. These are dynamic power and static power. Dynamic power refers to the power consumed during switching activities, while static power refers to leakage power consumed when the circuit is inactive [2], [16]. In Figure 9 the X-axis shows different type of SRAM cells and Y-axis power consumption. It is seen that ST10T cell in 32 nm CMOS technology contains the highest power consumption 13.77 μ W for both technologies and lowest value in ST12T which is 8.741 μ W. For CNFET technology ST10T SRAM cell shows the highest power 3.4222 μ W and lowest value provided by ST12T cell 1.5278 μ W.

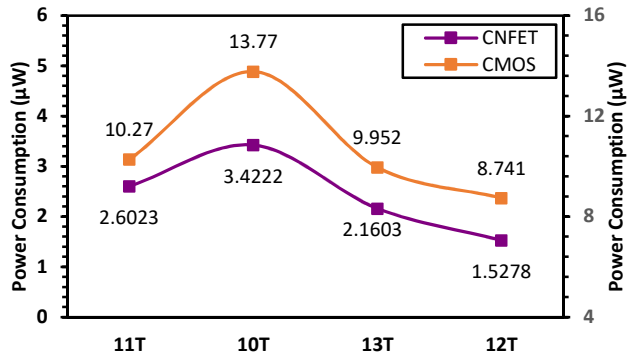


Fig. 9. Power consumption of different SRAM design using 32nm CMOS and CNFET technology.

B. Static power

The static power is defined as the power drawn from V_{DD} when the transistor is in hold mode which means all the switching operation of the SRAM are on hold and the data kept in the latch are same [7], [17]. According to the Figure 10 represented that for CMOS 32nm technology ST11T cell contains bigger static power 83.471 nW and ST13T cell lower for 7.045 nW. The biggest advantage of CNFET technology is that its static power is near-zero [18].

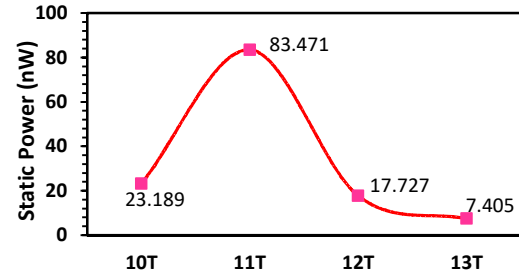


Fig. 10. Static power of different SRAM cells in 32nm CMOS technology.

C. Delay

It refers to the amount of time needed for an SRAM cell to complete a read or write operation immediately after the activation of the specific control signals. There are two kinds of delays; write delay and read delay. Write delay is defined as the time it takes Q or QB to reach 10% or 90% of V_{DD} after control signal is turned on [19]. On the other hand, read delay is the time it takes after control signal detects the change in BL or RBL and give a logical output accordingly [14]. Figure 11 and 12 shows read and write delay for CMOS and CNFET technology. In X-axis different SRAM cells are shown and Y-axis contains read and write delay. Figure 11 demonstrate that CMOS technology 'Write-1' is lowest value for ST10T cells 20.27 pS and 'Write-0' for ST12T cell 7.31 pS. And the minimum 'Read-1' is for ST13T cell 39.766 pS, 'Read-0' for ST11T cell 181.60 pS.

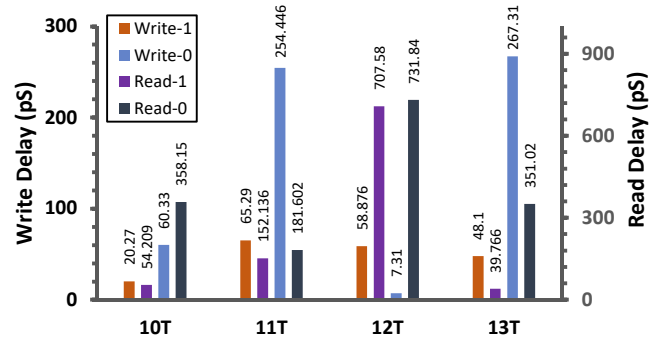


Fig. 11. WRITE and READ delay for CMOS technology.

For CNFET technology read, write delay is illustrated in Figure 12. ST11T shows lowest 'Write-1' 21.29 pS and 'Write-0' for ST10T cell 10.35 pS. For both 'Read-1' and 'Read-0' ST12T cell is minimum which is 29.2636 pS and 180.439 pS. In comparison, for CMOS write operation shows better results, whereas read operation performs better in CNFET.

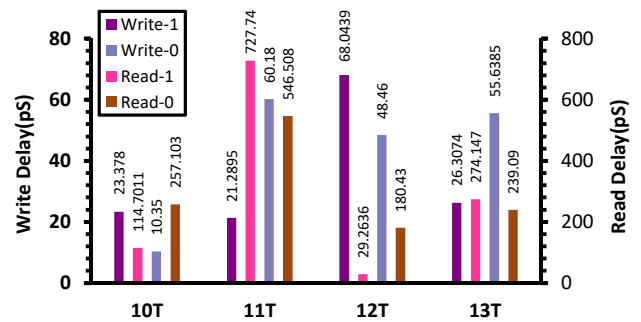


Fig. 12. WRITE and READ delay for CNFET technology.

D. Read SNM

The RSNM is defined as the maximum DC noise that a cell can tolerate without changing the data when read operation is performed [6]. The RSNM value was calculated from the upper lobe of the butterfly curve by measuring the maximum vertical length of the square fitted [16]. Figure 13 shows the RSNM of both technologies. For CMOS technology ST13T cell RSNM is maximum 375.76 mV and lowest is 271.43 mV for ST10T cell. Whereas for CNFET ST11T cell is highest 336.71 mV and lowest for ST12T cell 217.91 mV. Hence, it can be seen that read stability is much higher in CMOS technology as compared to CNFET. Figure 14 and Figure 15 shows butterfly curve of RSNM for CMOS and CNFET technology.

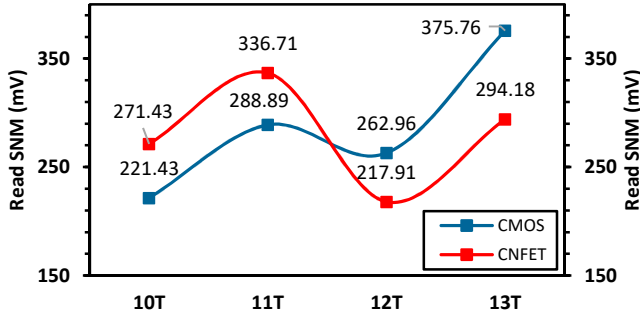


Fig. 13. SNM of READ for CMOS and CNFET technology.

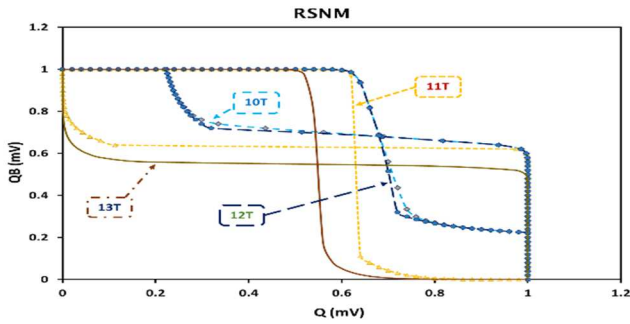


Fig. 14. RSNM of different SRAM design in CMOS.

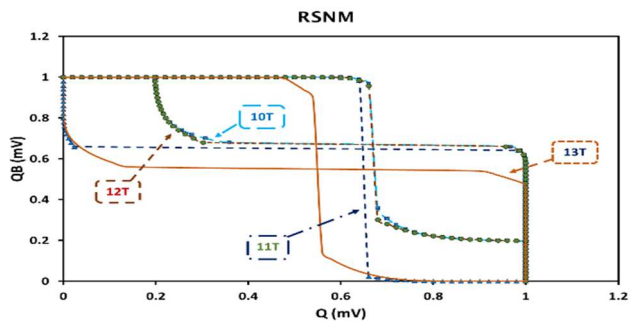


Fig. 15. RSNM of different SRAM design in CNFET.

E. Write SNM

It is defined as the minimum noise voltage that can flip the stored bit in a cell during write operation [2]. It is the measurement which says if the cell can still write in a noisy environment during write operation. In Figure 16 two technology WSNM data is shown. It is noticed that for CMOS the highest value is 640.35 mV for ST12T cell and minimum for ST13T cell 309.09 mV. In case for CNFET

ST12T is still greater 615.28 mV and ST10T is smaller for 371.83 mV. Write stability for CMOS shows more promising results compared to CNFET. Figure 17 and 18 shows the WSNM for CMOS and CNFET technology.

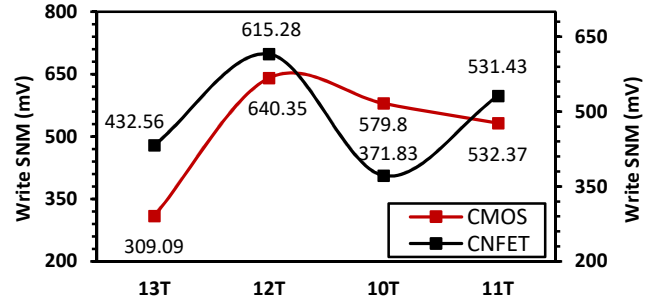


Fig. 16. SNM of WRITE for CMOS and CNFET technology.

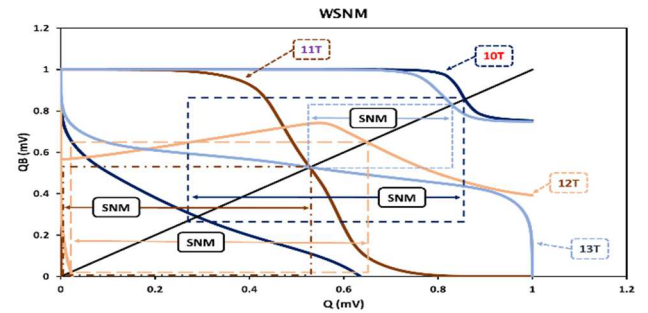


Fig. 17. WSNM of different SRAM design in CMOS.

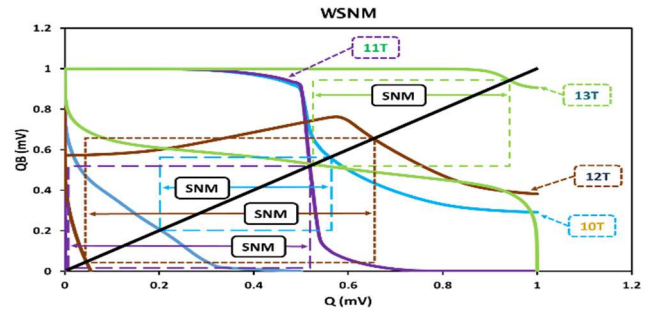


Fig. 18. WSNM of different SRAM design in CNFET.

F. Total Energy

The amount of energy consumed during read, write and hold operation in a memory cell [15], [16]. Figure 19 X-axis shows different cell and Y-axis shows energy consumed. The energy consumption of ST-based SRAM cells 10T, 11T, 12T, and 13T in CMOS technology is 1.3777 pJ, 1.027 pJ, 0.874 pJ, and 0.995 pJ, respectively. But, CNFET displays remarkable energy efficiency, consuming just 342.22 fJ, 260.23 fJ, 152.78 fJ, and 216.03 fJ and showing CNFET efficiency.

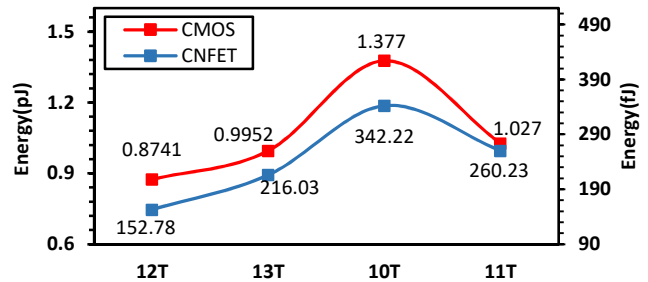


Fig. 19. Total energy of SRAM cells in CMOS and CNFET technology.

TABLE I. COMPARISON OF PARAMETERS ACROSS DIFFERENT SRAM CELLS AND TECHNOLOGIES

		Parameters							
		Power consumption (μ W)	Propagation Delay (pS)				SNM (mV)		Total Energy (pJ)
			<i>Write-1</i>	<i>Write-0</i>	<i>Read-1</i>	<i>Read-0</i>	<i>WSNM</i>	<i>RSNM</i>	
32nm CMOS	<i>10T</i>	13.77	20.27	60.33	54.21	358.15	579.80	221.43	1.377
	<i>11T</i>	10.27	65.29	254.45	152.14	181.60	532.37	288.89	1.027
	<i>12T</i>	8.74	58.88	7.31	707.58	731.84	640.35	262.96	0.874
	<i>13T</i>	9.95	48.10	267.31	39.77	351.02	309.09	375.76	0.995
32nm CNFET	<i>10T</i>	3.422	23.38	10.35	114.70	257.10	371.83	271.43	0.342
	<i>11T</i>	2.602	21.29	60.18	727.74	546.51	531.43	336.71	0.260
	<i>12T</i>	1.528	68.04	48.46	29.26	180.44	615.28	217.91	0.153
	<i>13T</i>	2.160	26.31	55.54	274.15	239.09	432.56	294.18	0.216

IV. CONCLUSION

The research that followed provided an in-depth analysis of ST based SRAM cell designs implemented in both CMOS and CNFET technologies. Additionally, it analyzes ST-based SRAM cells ST10T, ST11T, ST12T, and ST13T, evaluating their performance depending on several parameters including power consumption, delay, SNM (WSNM, RSNM), and total energy. The delay of 'Write-1' and 'Write-0' has demonstrated improved performance in 32 nm CMOS technology. Also, 'Read-1' and 'Read-0' CNFET revealed better outcomes. In comparison to CMOS technology, the power consumption of CNFET technology for 10T, 11T, 12T, and 13T have been decreased by 75.15%, 74.66%, 82.52%, 78.29%. The total energy consumption of ST10T, ST11T, ST12T, and ST13T cells decreases from 1.377 pJ, 1.027 pJ, 0.874 pJ, and 0.995 pJ in CMOS to 0.342 pJ, 0.260 pJ, 0.153 pJ, and 0.216 pJ in CNFET implementations. RSNM and WSNM CMOS technology have slightly improved performance compared to CNFET technology. Overall, the analysis reveals that ST12T cell implemented with CNFET technology delivers superior performance compared to its counterparts across both CMOS and CNFET platforms. This assessment revealed that CNFET technology performs more effectively in comparison to CMOS technology.

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